

Introductory Econometrics with Recitation — Quiz 3

April 14, 2026

Name: _____ ID: _____

1. **(17 points)** The following table reports estimates from regressions of delivery time, using a sample of 760 food-delivery orders collected from a delivery platform in 2025. The variables are defined as follows:

Variable Name	Description
Time	Total delivery time in minutes.
Distance	Distance from the restaurant to the customer (km).
Traffic	= 1 if the order was placed during a heavy-traffic period, 0 otherwise.
Rain	= 1 if it was raining at the time of delivery, 0 otherwise.
Bike	= 1 if the courier used a bicycle, 0 otherwise.
Downtown	= 1 if the destination is in Downtown, 0 otherwise.
Midtown	= 1 if the destination is in Midtown, 0 otherwise.
Uptown	= 1 if the destination is in Uptown, 0 otherwise.
Suburb	= 1 if the destination is in the Suburb, 0 otherwise.

Regressor	(1)	(2)	(3)
Distance	2.80	2.65	2.50
Traffic	9.40	8.90	8.60
Bike	-4.10	-3.60	-3.40
Rain		6.20	5.90
Downtown			7.10
Midtown			3.80
Uptown			5.20
Intercept	18.60	16.40	14.80
<i>SER</i>	12.50	11.80	10.20
<i>R</i> ²	0.227	0.312	0.487
<i>n</i>	760	760	760

- (a) **(3 points)** Using column (1), do orders placed during heavy-traffic periods take longer, on average, than orders placed outside heavy-traffic periods? By how much?
- (b) **(3 points)** Using column (2), predict the delivery time for an order with a distance of 4 kilometers, placed outside a heavy-traffic period, on a non-rainy day, and delivered by bicycle.
- (c) **(4 points)** Compute and interpret the adjusted R^2 for regression (3).

- (d) (**3 points**) Explain why the regressor *Suburb* is omitted from column (3).
- (e) (**4 points**) Interpret the meaning of *SER* for regression (1) in context. Then, holding the sample and the number of regressors fixed, explain how a decrease in *SSR* affects both *SER* and R^2 .

2. **(13 points)** Suppose that a researcher, using data on 410 walk-in customers and 190 customers with reservations at a popular restaurant, estimates the OLS regression:

$$\begin{array}{lcl} Wait = 42.30 - 13.70 \times Reserve, & R^2 = 0.31, & SER = 10.6 \\ (1.25) & (2.10) & \end{array}$$

where the numbers in parentheses are heteroskedasticity-robust standard errors for the regression coefficients.

The variable *Wait* is the waiting time, measured in minutes, and *Reserve* is a binary variable equal to 1 if the customer had a reservation and 0 if the customer was a walk-in.

- (a) **(4 points)** What is the estimated waiting-time gap between customers with reservations and walk-in customers? What is the average waiting time for each group in the sample?
- (b) **(5 points)** Let $\alpha = 0.05$. Is the estimated waiting-time gap significantly different from 0? Compute the p -value for testing the null hypothesis that having a reservation has no effect on waiting time.
- (c) **(4 points)** Do you think this regression suffers from omitted variable bias? Explain why. Which variables would you add to the regression?